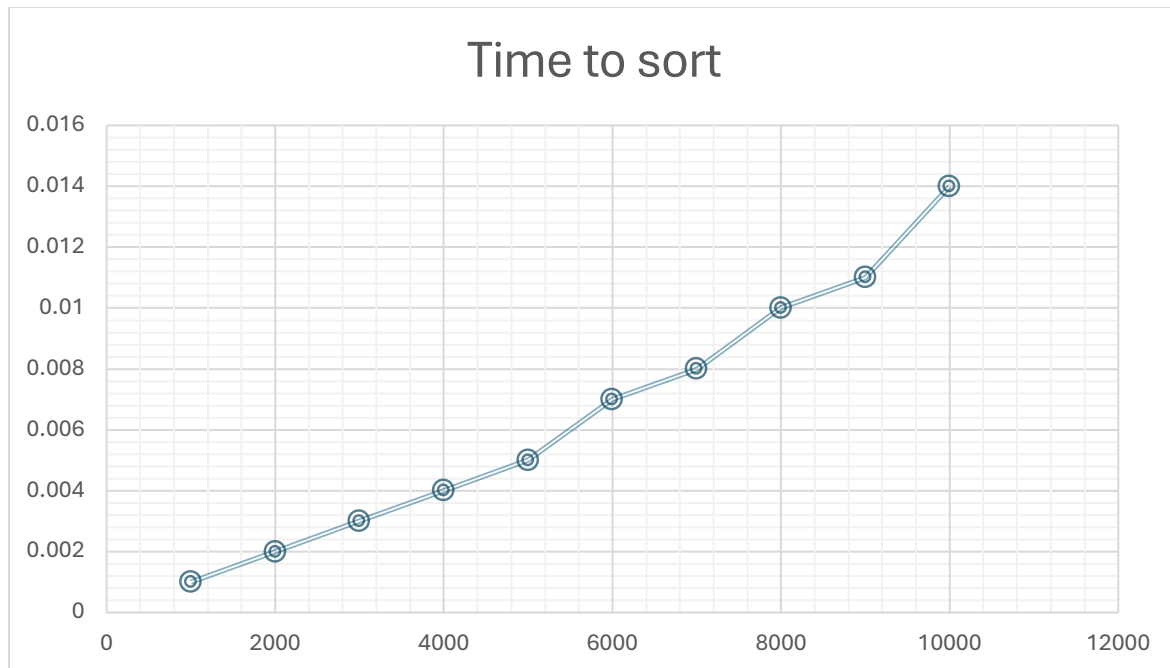




Size of Array	Time to sort
1000	0.001
2000	0.002
3000	0.003
4000	0.004
5000	0.005
6000	0.007
7000	0.008
8000	0.01
9000	0.011
10000	0.014

The big O of this quick sort algorithm is $O(n \log n)$. Because the algorithm splits the array in half and sorts each half recursively.

The C Profiler shows $n \log n$ time on the graph in visual terms. The time it took to sort the array grew as the size of the array grew. I noticed the time to sort and the amount of elements in the arrays were sync. For example 1000 elements took 0.001 seconds, 2000 took 0.002 second and up to 5000 elements it kept that pattern. At 6000 elements it changed and then a new pattern showed up as $n \log n$ times started to show its self-more definitively.

```
jjohnson1 python JohnsonJeffMod1PA.py
2432 function calls (814 primitive calls) in 0.001 seconds
```

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.001	0.001	<string>:1(<module>)
1619/1	0.000	0.000	0.001	0.001	JohnsonJeffMod1PA.py:2(quickSort)
809	0.000	0.000	0.000	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.001	0.001	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

```
jjohnson1 python JohnsonJeffMod1PA.py
5405 function calls (1805 primitive calls) in 0.002 seconds
```

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.002	0.002	<string>:1(<module>)
3601/1	0.001	0.000	0.002	0.002	JohnsonJeffMod1PA.py:2(quickSort)
1800	0.001	0.000	0.001	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.002	0.002	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

```
jjohnson1 python JohnsonJeffMod1PA.py
8402 function calls (2804 primitive calls) in 0.003 seconds
```

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.003	0.003	<string>:1(<module>)
5599/1	0.001	0.000	0.003	0.003	JohnsonJeffMod1PA.py:2(quickSort)
2799	0.002	0.000	0.002	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.003	0.003	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

```
jjohnson1 python JohnsonJeffMod1PA.py
11402 function calls (3804 primitive calls) in 0.004 seconds
```

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.004	0.004	<string>:1(<module>)
7599/1	0.001	0.000	0.004	0.004	JohnsonJeffMod1PA.py:2(quickSort)
3799	0.003	0.000	0.003	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.004	0.004	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

jjohnson1 python JohnsonJeffMod1PA.py
14402 function calls (4804 primitive calls) in 0.005 seconds

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.005	0.005	<string>:1(<module>)
9599/1	0.002	0.000	0.005	0.005	JohnsonJeffMod1PA.py:2(quickSort)
4799	0.003	0.000	0.003	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.005	0.005	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

jjohnson1 python JohnsonJeffMod1PA.py
17402 function calls (5804 primitive calls) in 0.007 seconds

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.007	0.007	<string>:1(<module>)
11599/1	0.002	0.000	0.007	0.007	JohnsonJeffMod1PA.py:2(quickSort)
5799	0.005	0.000	0.005	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.007	0.007	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

jjohnson1 python JohnsonJeffMod1PA.py
20402 function calls (6804 primitive calls) in 0.008 seconds

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.008	0.008	<string>:1(<module>)
13599/1	0.002	0.000	0.008	0.008	JohnsonJeffMod1PA.py:2(quickSort)
6799	0.006	0.000	0.006	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.008	0.008	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

jjohnson1 python JohnsonJeffMod1PA.py
23402 function calls (7804 primitive calls) in 0.010 seconds

Ordered by: standard name

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.010	0.010	<string>:1(<module>)
15599/1	0.003	0.000	0.010	0.010	JohnsonJeffMod1PA.py:2(quickSort)
7799	0.007	0.000	0.007	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.010	0.010	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

jjohnson1 python JohnsonJeffMod1PA.py
26402 function calls (8804 primitive calls) in 0.011 seconds

Ordered by: standard name

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.011	0.011	<string>:1(<module>)
17599/1	0.003	0.000	0.011	0.011	JohnsonJeffMod1PA.py:2(quickSort)
8799	0.008	0.000	0.008	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.011	0.011	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}

jjohnson1 python JohnsonJeffMod1PA.py
29402 function calls (9804 primitive calls) in 0.014 seconds

Ordered by: standard name

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	0.014	0.014	<string>:1(<module>)
19599/1	0.004	0.000	0.014	0.014	JohnsonJeffMod1PA.py:2(quickSort)
9799	0.010	0.000	0.010	0.000	JohnsonJeffMod1PA.py:8(lumuto)
1	0.000	0.000	0.014	0.014	{built-in method builtins.exec}
1	0.000	0.000	0.000	0.000	{built-in method builtins.len}
1	0.000	0.000	0.000	0.000	{method 'disable' of '_lsprof.Profiler' objects}